QUANT

Number System

(Digital Root, Unit's Digit, Last two digits
& other important concepts)

Assignment 01

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fNumbers Online Assignment -01

1. A hundred digit number is formed by writing first 54 natural numbers in front of each other as 123456…….5354. Find the remainder when this number is divided by 8.

1. 4 2. 7 3. 2 4. 0

2. If $n = 1 + x$, where x is the product of four consecutive positive integers, then which of the following statements is/ are true?

(1) n is odd (2) n is prime (3) n is a perfect square

1. 1 only 2. 2 only 3. 3 only 4. 1&3 only

3. A certain number when divided by 899 gives a remainder 63. What is the remainder when the same number is divided by 29?

1. 25 2. 5 3. 27 4. Data inadequate

4. If N = 1421 × 1423 × 1425, what is the remainder when 'N' is divided by 12?

1. 0 2. 1 3. 3 4. 9

5. A papa number is defined as the corresponding single digit number obtained by successive addition of digits of the original number till we finally arrive at a single digit.

E.g.:- Papa number of 1489 = 1 + 4 + 8 + 9 = 22 = 2 + 2 = 4

Which of the following numbers is completely divisible by its Papa number?

1. 5555 2. 3254 3. 6666 4. 7071

6. As defined in the above question, how many two-digit prime numbers have their papa numbers as prime again?

1. 9 2. 10 3. 11 4. 12

7. For how many values of a are, a, $a + 4$, $a + 7$ prime numbers?

1. One 2. Two 3. None 4. Infinite

8. For any positive integer z, $z^5 + z^3 + 3$ is always

1. odd 2. even 3. either odd or even 4. Data inadequate

9. A four-digit even number $abcd$ is divisible by 55, which of the following could be the value of a, b and c respectively?

1. 3, 0, 8 2. 9, 5, 4 3. 7, 4, 3 4. 5, 0, 5

10. Find the smallest 6-digit number which is a multiple of 17.

1. 100006 2. 100011 3. 100013 4. None of these

Answer keys & Explanations

Numbers Online Assignment 01		
1.	3	We need to look at only the last three digits of this number. So 354 divided by 8 gives remainder as 2. Thus 2 is the answer. Thus 3rd option.
2.	4	Assume values of *x* to get the answer. We can find that 1st and 3rd statements are always true. So answer is 4th option.
3.	2	The number can be written as 899x + 63. Now this has to be divided by 29. Now 899 = 31 □ 29, so it is an exact multiple of 29. So the first term will leave a remainder of 0. 63 will leave a remainder of 63 – 58 = 5.
4.	3	N = 1421 × 1423 × 1425. Remainders when these numbers are divided by 12 are 5, 7and 9. Their product is 315. Divide it by 12 and find the remainder to be 3.
5.	3	Going by options, we get the answer as 3rd option as Papa number of 6666 is 6 + 6 + 6 + 6 = 24 ⇒ 2 + 4 = 6. So we can see that 6666 is divisible by 6.
6.	3	There are 11 prime numbers i.e. 11, 23, 29, 41, 43, 47, 59, 61, 79, 83, 97 which are having their Papa numbers as prime numbers again.
7.	3	No value satisfies the given condition as for a + 7 to be prime, *a* has to be even. As a result *a* + 4 will never be prime.
8.	1	Whether z is even or odd $z^5 + z^3$ will always be even. When 3 is added to this the result will be odd. Thus odd will be answer.
9.	1	55 = 11 × 5. The unit's digit must be either 0 or 5. Then apply the divisibility of 11 and see the first option could be the value of a, b and c respectively. Here we are assuming the d to be 0. But the question is 'could be' and only first option could be the answer.
10.	2	Take the smallest six digit no. divide by 17 and get the remainder as 6. Now if u add six the remainder will become 6 + 6 = 12. Think it as, the smallest no. that should be added in 6 to make it a multiple of 17 i.e. 17 – 6 = 11. Hence the smallest such no. is 100000 + 11= 100011.